

Zihan (Joe) Yang

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Education

University of Illinois Urbana-Champaign, Champaign, United States
Bachelor of Science in Materials Science and Engineering
Bachelor of Science in Statistics
Minors: Semiconductor Engineering; Computational Science and Engineering
Grainger College of Engineering Edmund J. James Scholar Honor Student

August 2023 — May 2027
GPA: 3.99 / 4.00
Dual Degree Candidate

Research Interests

Electrochemical Energy Conversion with Autonomous Materials Discovery, Computational Simulations, and Data-Driven Modeling.

Research Experience

Hong Yang's Group, Department of Chemical and Biomolecular Engineering
Undergraduate Research Assistant

Supervisor: Prof. Hong Yang
August 2024 — Present

- 1. Data-Driven Modeling for Ultrathin Gold Nanowire Kinetics Growth Behaviors via Statistical Learning.**
 - I. Modeled gold nanowire length distributions using statistical modeling using TEM experimental characteristic lengths from length–time datasets across various synthetic conditions.
 - II. Developed and fit a mechanistic rate-equation model via numerical integration to extract kinetic constants across conditions and connect statistics to microscopy informed growth mechanisms.
- 2. Syngas-Driven Composition Control for Ultrathin Gold Nanowire Synthesis.**
 - I. Investigated H₂/CO gas composition effects on Au precursor reduction and AuNP byproduct formation using in situ UV–vis analysis.
 - II. Quantified TEM-derived AuNW selectivity across gas ratios to identify a syngas window that promotes wire-dominant growth while suppressing nanoparticle formation.
- 3. Ferrocene-Functionalized Polymer with Sulfonated Binders for Acidic Oxygen Evolution Reaction.**
 - I. Formulated Y₂Ru₂O_{7- δ} catalyst inks with binder-chemistry tuning to improve dispersion and acidic stability.
 - II. Demonstrated ink stability differences under acidic electrochemical operations and quantified oxidative durability and fluoride loss using chemical stress tests with fluoride-selective measurements.

Federal Institute for Materials Research and Testing (BAM)
Bundesanstalt für Materialforschung und -prüfung, Berlin, Germany
Summer Research Assistant, Division 6.2: Materials and Surface Technologies

Supervisor: Dr.-Ing. Özlem Özcan Sandikcioglu
May 2026 — Present

- 1. Project: Development of electrochemical corrosion testing workflows in an autonomous laboratory (self-driving-lab/material acceleration platform) with the application of energy conversion and corrosion.**

Publications and Manuscripts

- Yan-Ru Chen, **Zihan (Joe) Yang**, Binyao Qi, Anran Song, Hong Yang. Polydispersity Index Analysis of Living Synthesis of High Aspect-Ratio, Ultrafine Gold Nanowire. To be Submitted.
- Yan-Ru Chen, Gavin Hazen, **Zihan (Joe) Yang**, Jiaxuan (Estella) Wang, Johannes Elbert, Yongqi Lu, Xiao Su, Joaquin Lopez, Hong Yang. Scalable synthesis of gold nanowires via syngas-mediated kinetic control for Sustainable Rare Earth Recovery. Work in progress.
- Yan-Ru Chen, Johannes Elbert, Elizabeth Salako, Matthew Thompson, **Zihan (Joe) Yang**, Yong Gao, Xiao Su, Hong Yang. Homogenizing Poly(perfluorosulfonic acid)-Catalyst Ink Formulation with Ferrocene-based Polymer for Acidic OER. Manuscript in preparation.

Teaching Experience

Undergraduate Teaching Assistant for MSE 206: Mechanics for MatSE
Class Size: 99

Supervisor: Prof. Wenjie Zhou
February 2026 — May 2026

- Designed and led multiple discussion sections, provided supplemental lectures, facilitated real-world problem-solving sessions, and provided 1:1 support to students via office hours and report grading.
- Managed course assessments and debugged in PrairieLearn, addressed online questions submitted via Campuswire, and organized course materials on Canvas.

Academic Projects

Operando Analysis of Li Anode–Solid Electrolyte Interface Evolution

Proposal Contributor

Supervisor: Prof. Mei Shen

April 2026 — May 2026

1. Proposed an operando electrochemical validation framework combining SEM/X-ray CT, voltage profiles, and EIS to track Li/SE interface degradation.
2. Designed a high-throughput PFM/ML workflow to optimize interfacial morphology, transport, kinetics, and mechanical descriptors for stable Li stripping/plating.

Coursework Development: Atomic Scale Simulations Digitized Collection

Content Contributor

Supervisor: Prof. Antonia Statt

October 2025 — December 2025

1. Authored and edited textbook chapters on Molecular Dynamics (MD) thermostats and Langevin dynamics, connecting theory, derivations, and implementation.
2. Built reproducible workflows linking numerical stability, sampling behavior, and physical interpretation in MD simulations.

Molecular Dynamics Simulation of Defect Diffusion in Si Annealing

Project Team Member

Supervisor: Prof. Antonia Statt

September 2025 — December 2025

1. Built LAMMPS Molecular Dynamics workflows to simulate defect diffusion in crystalline Si with controlled defect and carbon concentrations.
2. Computed and analyzed defected Si structure, extracted diffusion coefficients via hopping with Arrhenius analysis, and automated sweeps for post-processing data analysis.

Selected Coursework

Materials Science and Engineering

1. CHEM 524: Electrochemical Methods
2. MSE 498: Computational Thermodynamics and Kinetics
3. MSE 485: Atomic Scale Simulations
4. MSE 480: Surfaces and Colloids

Statistics

1. STAT 432: Statistical Learning
2. STAT 431: Applied Bayesian Analysis
3. STAT 428: Statistical Computing
4. STAT 425/426: Statistical Modeling

Awards and Honors

Grainger Continuing Student Scholarship

May 2026

Materials Science and Engineering BAM Summer Research Scholarship

May 2026

Materials Science and Engineering Henry E. Grein Jr. Scholarship

April 2026

Materials Science and Engineering Henry E. Grein Jr. Scholarship

April 2025

Samsung Semiconductor Technology Track Scholarship

August 2024

University of Illinois Dean's List (6 Semesters)

Fall 2023 — Present

Leadership and Service

Make Your Own World DIY Club

Founder, Membership Director

Registered School Organizations

August 2024 — Present

1. Developed outreach programs, attracted 300+ new members and retained existing members.
2. Managed the club's social media accounts to support outreach and promote monthly club events.
3. Led efforts to secure recognition by MatSE department and developed 3D printing platform for club members.

Nanjing Collaborators

College Student Volunteer

Nanjing, China

August 2021 — Present

1. Lead weekly engineering workshops for underrepresented middle school students on energy and environment.
2. Develop concise lesson materials and real-world examples to support student learning.

Skills

- **Programming:** Python; R; Java.
- **Computational Tools:** CALPHAD, MD (LAMMPS), MC, DFT, OOF2.
- **Experiment & Characterization:** Sol-gel Synthesis; Electrochemistry (CV, LSV); OM; XRD; SEM; TEM; FTIR.
- **Languages:** English (Professional), Mandarin (Native), German (Elementary).